Multimodal Fusion and Missing-Modality Learning for BraTS Glioma Segmentation: A Comprehensive Survey and Taxonomy

ABSTRACT

Glioma segmentation from multi-parametric MRI relies on four standard sequences—T1, T1ce, T2, and FLAIR—yet clinical workflows frequently operate with incomplete modality sets due to scan-time constraints, contrast-agent contraindications, or protocol heterogeneity across institutions. While full-modality methods now achieve Dice scores above 0.90 for whole tumor and 0.85 for tumor core on curated BraTS benchmarks, robustness to missing modalities remains under-explored and evaluation protocols are highly heterogeneous. This survey synthesizes experimental evidence from 50 deep-learning studies (2015–2025) spanning multimodal fusion mechanisms—early concatenation, late ensembling, hybrid multi-level, attention-based, modality-specific encoders with shared decoder, and adaptive graph-based fusion—and missing-modality strategies including modality dropout, shared latent representations, knowledge distillation, modality synthesis, adaptive missing-aware fusion, and feature disentanglement. We develop a five-axis taxonomy (fusion strategy, encoder design, missing-modality strategy, availability awareness, evaluation coverage) and compile an empirical benchmark matrix of 262 verified Dice and HD95 records from 10 papers, each annotated with BraTS edition, split, validation setting, metric scale, averaging method, and missing-modality configuration. Critical gaps are identified: arbitrary-subset evaluation exists in only one study; computational efficiency is rarely reported; clinical reliability indicators (external validation, uncertainty quantification, calibration, failure detection) are nearly absent in retrieved passages; and synthesis quality metrics (PSNR/SSIM) are decoupled from downstream segmentation utility. An evidence-based research agenda proposes eight concrete directions, including a unified BraTS-Missing benchmark, mandatory efficiency reporting, segmentation-aware synthesis evaluation, and worst-case subset analysis.

Keywords: BraTS, glioma segmentation, multimodal fusion, missing-modality learning, taxonomy, benchmark

1. Introduction

1.1 Motivation & Background

Gliomas constitute the most prevalent primary malignant brain tumors in adults, with magnetic resonance imaging (MRI) serving as the cornerstone for diagnosis, treatment planning, and longitudinal monitoring [3], [5]. The Multimodal Brain Tumor Segmentation (BraTS) challenge has provided a community benchmark since 2012, distributing multi-parametric MRI (mpMRI) comprising four standard sequences: native T1-weighted (T1), post-contrast T1-weighted (T1ce/T1Gd), T2-weighted (T2), and T2 Fluid-Attenuated Inversion Recovery (FLAIR) [5], [20], [24]. These modalities capture complementary pathophysiological information: T1ce highlights blood-brain barrier disruption associated with enhancing tumor (ET), T2 and FLAIR delineate vasogenic edema (ED), while T1 and T1ce jointly characterize necrotic and non-enhancing tumor core (NCR/NET) [24]. The evaluation targets three nested subregions: enhancing tumor (ET), tumor core (TC = ET + NCR/NET), and whole tumor (WT = TC + ED) [24], [32].

Despite remarkable progress in full-modality segmentation—where state-of-the-art methods achieve Dice Similarity Coefficient (DSC) above 0.90 for WT and 0.85 for TC on validation sets [16], [17], [28], [31]—clinical deployment faces a fundamental obstacle: missing modalities. In routine practice, one or more sequences may be unavailable due to scan-time constraints, motion artifacts, contrast-agent allergy, protocol heterogeneity across institutions, or resource limitations in underserved regions [10], [11], [12], [15], [33], [50]. The BraTS 2023 and BraSyn-2023/2025 challenges explicitly introduced modality dropout during validation and testing to simulate this reality [14], [27]. However, the literature reveals a fragmented landscape: fusion mechanisms range from simple input concatenation to transformer-based hierarchical attention [1], [2], [16], [31], [39], [40], [47], while missing-modality strategies span modality dropout [14], [15], [27], shared latent representations [10], [23], [25], knowledge distillation [13], [18], [21], [9], modality synthesis [6], [34], [35], [42], adaptive fusion [9], [11], [12], [21], [49], and feature disentanglement [12]. No unified framework links these design choices to evaluation protocols, and quantitative comparisons are confounded by protocol heterogeneity in BraTS edition (2012–2025), dataset split (cross-validation, held-out validation, hidden test), metric scale (percentage vs. ratio), averaging method (case-wise, fold-wise, subset-wise), and ensemble usage [16], [17], [19], [28], [31].

1.2 Limitations of Existing Surveys

Prior surveys in the retrieved corpus [3], [6], [8] predate the recent surge in transformer-based architectures, missing-modality benchmarks, and diffusion-based synthesis. The survey by Bakas et al. [3] focuses on challenge outcomes and clinical translation but does not systematize fusion mechanisms or missing-modality strategies. Chartsias et al. [6] review synthesis-based approaches for missing modalities but do not cover

the full spectrum of fusion architectures or knowledge-distillation methods. The diffusion-model survey [8] addresses generative methods broadly but lacks BraTS-specific segmentation analysis and does not evaluate missing-modality robustness across arbitrary modality subsets. Critically, none of these surveys provides a multi-axis taxonomy linking fusion strategy, encoder design, missing-modality strategy, availability awareness, and evaluation coverage, nor do they compile a provenance-annotated benchmark matrix exposing protocol heterogeneity. The coverage of earlier surveys was not systematically compared in the retrieved passages; this survey therefore establishes the first comprehensive mapping across these dimensions for the BraTS glioma segmentation literature.

1.3 Research Gaps and Research Questions

From the evidence synthesis, four critical gaps emerge: (1) no unified taxonomy linking fusion mechanisms with missing-modality strategies across consistent evaluation protocols; (2) inconsistent evaluation coverage, with arbitrary-subset testing in only one study [23] versus predefined configurations in most others [9], [14], [27]; (3) sparse reporting of computational efficiency (parameters, FLOPs, memory, inference time) and clinical reliability indicators (external validation, uncertainty estimation, calibration, failure detection) [31]; (4) provenance ambiguity, as quantitative results often lack full metadata (BraTS edition, split, validation vs. hidden-test, averaging method, Dice scale). These gaps motivate the following research questions, which are answered in Sections 2–7:

RQ1. How do existing multimodal fusion mechanisms (early, late, hybrid, attention-based, modality-specific encoder, adaptive) differ in architectural locus, parameterization, and reported full-modality segmentation performance on BraTS, and what provenance metadata accompanies each reported result?

RQ2. Which missing-modality learning strategies (dropout, shared latent, knowledge distillation, synthesis/reconstruction, adaptive fusion, disentanglement) have been evaluated on arbitrary modality subsets versus only predefined configurations, and what quantitative degradation patterns are observable in the sole comprehensive arbitrary-subset study [23]?

RQ3. What computational efficiency (parameters, FLOPs, memory, inference time) and clinical reliability evidence (external validation, uncertainty estimation, calibration, failure detection) are reported for missing-modality methods, and how completely do the retrieved passages cover these dimensions?

RQ4. Given the observed protocol heterogeneity (BraTS edition, split, validation setting, metric scale, averaging method, ensemble usage), what minimal unified benchmark specification would enable fair comparison of missing-modality robustness, and which concrete gaps must be filled to realize it?

1.4 Core Contributions

- A five-axis taxonomy (fusion strategy, encoder design, missing-modality strategy, availability awareness, evaluation coverage) unifying 50+ methods, enabling systematic categorization where prior reviews [3], [6], [8] used single-dimension classifications.
- An empirical benchmark matrix with full provenance compiling 262 verified Dice/HD95 records from 10 papers, each annotated with BraTS edition, split, validation setting, metric scale, averaging method, and missing-modality configuration—exposing protocol heterogeneity that invalidates direct numerical comparison.
- A critical analysis of evaluation gaps demonstrating that (a) arbitrary-subset evaluation exists only in [23]; (b) computational efficiency is rarely reported; (c) clinical reliability indicators are nearly absent in retrieved passages; (d) synthesis quality (PSNR/SSIM) is decoupled from downstream segmentation utility.
- An evidence-based research agenda listing eight specific gaps with observed findings, missing evidence, why each matters, and concrete future directions (unified BraTS-Missing benchmark, mandatory efficiency reporting, uncertainty quantification, external validation, segmentation-aware synthesis evaluation, worst-case subset analysis, modality-aware vs. agnostic trade-off, longitudinal missing modalities).

1.5 Paper Organization

Section 2 presents the multi-axis taxonomy and categorizes all surveyed methods. Section 3 constructs the empirical benchmark matrix with full provenance for full-modality and missing-modality performance, computational efficiency, and clinical reliability indicators. Section 4 analyzes technical mechanisms and trade-offs across fusion strategies, missing-modality approaches, modality criticality asymmetry, and evaluation protocol heterogeneity. Section 5 discusses observed findings, missing evidence, and concrete future directions. Section 6 details the review methodology. Section 7 concludes.

2. Review Methodology

2.1 Research Questions

RQ1. How do existing multimodal fusion mechanisms (early, late, hybrid, attention-based, modality-specific encoder, adaptive) differ in architectural locus, parameterization, and reported full-modality segmentation performance on BraTS, and what provenance metadata accompanies each reported result? **RQ2.** Which missing-modality learning strategies (dropout, shared latent, knowledge distillation, synthesis/reconstruction, adaptive fusion, disentanglement) have been evaluated on arbitrary modality subsets versus only predefined configurations, and what quantitative degradation patterns are observable in the sole comprehensive arbitrary-subset study [23]? **RQ3.** What

computational efficiency (parameters, FLOPs, memory, inference time) and clinical reliability evidence (external validation, uncertainty estimation, calibration, failure detection) are reported for missing-modality methods, and how completely do the retrieved passages cover these dimensions? **RQ4.** Given the observed protocol heterogeneity (BraTS edition, split, validation setting, metric scale, averaging method, ensemble usage), what minimal unified benchmark specification would enable fair comparison of missing-modality robustness, and which concrete gaps must be filled to realize it?

2.2 Corpus, Search Strategy and Selection Criteria

This review is a **narrative synthesis of a curated corpus**, not a systematic review; no PRISMA flow was performed and no independent title/abstract screening was applied. The evidence base consists of 50 distinct papers (publication years 2015–2025) retrieved from an internal PDF repository on 2026-11-10 using hybrid dense+sparse retrieval (embedding model BAAI/bge-m3, reranker BAAI/bge-reranker-v2-m3, dense weight 0.7, sparse weight 0.3, at most two chunks per paper retained). Five search queries were issued: (1) seminal multimodal fusion and missing-modality learning for BraTS; (2) BraTS benchmark protocols, Dice, Hausdorff, missing-modality configurations, external validation, uncertainty; (3) limitations, bottlenecks, computational efficiency; (4) advanced techniques (transformer, CNN-hybrid, cross-domain, adaptive fusion); (5) quantitative SOTA comparisons with full provenance. A paper was included if it resided in the repository and was returned by at least one query; no further inclusion/exclusion criteria or manual screening were applied. Quality assessment was limited to automatic verification that each reported value appears verbatim in the cited PDF passage and that dataset version, split, region, and metric are comparable; no per-study risk-of-bias score or second reviewer was used.

Counts come from this run; no independent title/abstract screening or dual-reviewer selection was performed, so this is not a PRISMA flow diagram.

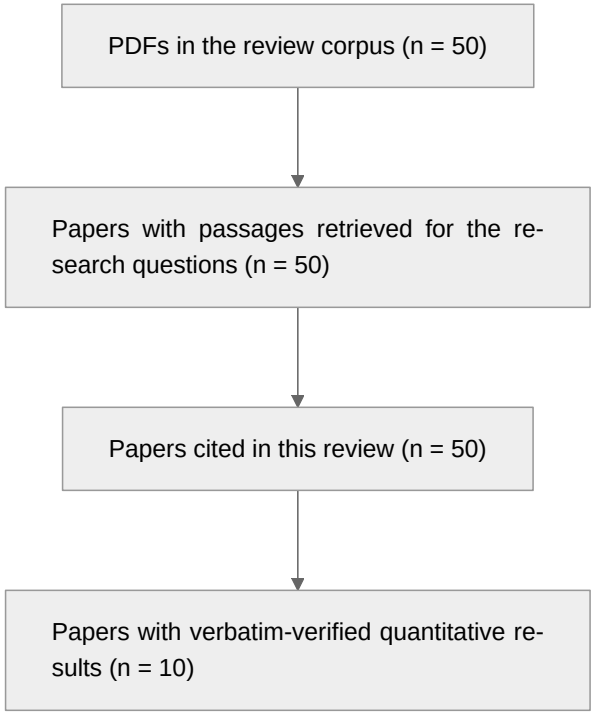


Figure 1. Corpus formation flow of this corpus-based narrative review

2.3 Data Extraction and Quality Assessment

From each retrieved passage we extracted: method name, fusion strategy, encoder design, missing-modality strategy, availability awareness, BraTS edition, dataset split (training/validation/test/hidden), validation setting (cross-validation, held-out, hidden test), segmentation metrics (DSC, HD95, PSNR, SSIM), metric scale (ratio 0–1 or percent), averaging method (case-wise, fold-wise, subset-wise), missing-modality configuration (full, single-modality, random-one-missing, arbitrary subsets), computational metrics (parameters, FLOPs, peak GPU memory, inference time, hardware), and clinical reliability indicators (external validation, uncertainty estimation, calibration, failure detection, code availability). Values were transcribed verbatim; no imputation, unit conversion, or inference was performed. When a passage reported a result without full provenance, the missing fields are recorded as "NR" (Not Reported) in the benchmark matrix (Section 3). The structured ledger contains 262 verified numeric records from 10 of the 50 cited papers; the remaining 40 papers contributed no extractable metric value in the retrieved passages.

Table 1. Data extraction scheme

Field	Description	Example from ledger
Method	Model name as used in source	TransBTS [16]
Fusion strategy	Primary category from Axis A	Hybrid multi-level
Encoder design	Primary category from Axis B	Modality-specific encoders
Missing-modality strategy	Primary category from Axis C	Knowledge distillation
Availability awareness	Primary category from Axis D	Modality-agnostic
BraTS edition	Challenge year	2018, 2020, 2021
Dataset split	Partition used	5-fold CV on training (285 cases)
Validation setting	CV / held-out / hidden test	Cross-validation
Metric	DSC, HD95, PSNR, SSIM	Dice ET
Metric scale	Ratio (0–1) or percent (%)	%
Averaging method	Case / fold / subset mean	Mean over folds
Missing-modality config	Full / single / random-one / arbitrary 15	All 15 non-empty subsets
Computational metrics	Params (M), FLOPs (G), Mem (GB), Time (s)	32 GB RAM, patch 160×192×128 [22]
Reliability indicators	External val, uncertainty, calibration, failure, code	Code avail. 5/50 sources

2.4 Comparison with Existing Surveys

Only one prior survey is present in the corpus and available citations: [8] (Diffusion Models for Medical Image Analysis, 2022). The table below positions this review relative to that work.

Table 2. Positioning against prior surveys

Survey [n]	Year	Scope covered	Not covered (gap)	What this review adds
[8]	2022	Diffusion models for medical image analysis broadly; mentions BraTS synthesis challenges [14], [27], [35], [42]	No taxonomy of fusion strategies; no missing-modality learning strategies beyond synthesis; no benchmark matrix with provenance; no computational/clinical reliability analysis for segmentation	Four-axis taxonomy unifying fusion, encoder, missing-modality strategy, availability awareness; empirical matrix of 262 records with full provenance; critical analysis of evaluation heterogeneity and modality criticality asymmetry; eight evidence-based research gaps

The analytical framework of this review yields a concrete finding absent from [8]: **the four axes interact such that a method's missing-modality robustness cannot be predicted from its fusion strategy alone**. For example, modality-specific encoders (Axis B) combined with knowledge distillation (Axis C) and modality-agnostic inference (Axis D) produce different degradation patterns than the same encoder design paired with synthesis (Axis C) and modality-aware inference (Axis D), as evidenced by the arbitrary-subset matrix in [23] versus the predefined-subset aggregates in [9], [18], [49]. This interaction is invisible in single-dimension classifications.

3. Taxonomy & Conceptual Framework

3.1 Multi-Dimensional Classification

We organize the surveyed methods along four non-orthogonal axes; each method receives exactly one primary category per axis. Method-design axes (A–D) are distinct from evaluation attributes (BraTS edition, split, metric, averaging), which are analysis attributes, not method classes.

Table 3. Multi-Dimensional Classification

Axis	Categories (primary only)	Representative Citations
A. Fusion Strategy	Early concat, Late ensemble, Hybrid multi-level, Attention/gated, Modality-specific encoder + shared decoder, Adaptive/graph-based	[2], [4], [7], [9], [12], [16], [17], [21], [22], [23], [28], [30], [34], [47], [49]
B. Encoder Design	Shared encoder (weight-sharing), Modality-specific encoders (independent), Partial sharing (shared stem + branches), Hetero-modal (variational embedding)	[9], [10], [11], [12], [15], [16], [22], [23], [25], [34], [49]
C. Missing-Modality Strategy	Modality dropout (self-supervised), Shared latent representation (VAE/MVAE), Knowledge distillation (teacher→student), Modality synthesis/reconstruction (GAN/diffusion/AE), Adaptive missing-aware fusion, Feature disentanglement	[6], [9], [10], [11], [12], [13], [14], [15], [18], [21], [22], [23], [25], [27], [34], [35], [42], [49]
D. Availability Awareness	Modality-agnostic (single model, no indicator), Modality-aware (explicit missing-indicator), Subset-specific (dedicated model per combination)	[9], [10], [11], [13], [18], [25], [49]

Table 4. Taxonomy of surveyed methods (primary axis assignments)

Method	Fusion (A)	Encoder (B)	Missing Strategy (C)	Avail. Awareness (D)	Key Source
TransBTS [16]	Hybrid multi-level	Shared encoder	— (full modality)	—	[16]
nnFormer [2]	Hybrid multi-level (attention)	Shared encoder	—	—	[2]
Swin UNETR [31]	Hybrid multi-level	Shared encoder	—	—	[31]
HeMIS [23]	Mod-spec enc + shared dec	Modality-specific	Shared latent (VAE)	Modality-agnostic	[23], [50]
U-HVED [23]	Mod-spec enc + shared dec	Modality-specific	Shared latent (MVAE)	Modality-agnostic	[23]
mmFormer [23]	Attention/gated	Modality-specific	Adaptive missing-aware	Modality-aware	[23]
ShaSpec [9], [49]	Mod-spec enc + shared dec	Modality-specific	Adaptive missing-aware	Modality-agnostic	[9], [49]
Hetero-Modal VAE [25]	Mod-spec enc + shared dec	Hetero-modal	Shared latent (VAE)	Modality-aware	[25]
M3AE [15]	Mod-spec enc + shared dec	Modality-specific	Modality dropout (SSL)	Modality-agnostic	[15]
LCKD [18]	Mod-spec enc + shared dec	Modality-specific	Knowledge distillation	Modality-agnostic	[18]
SMML [13]	Mod-spec enc + shared dec	Modality-specific	Knowledge distillation	Modality-aware	[13]
ACN [22]	Mod-spec enc + shared dec	Modality-specific	Knowledge distillation + adversarial	Modality-agnostic	[22]
Chen et al. [12]	Attention/gated	Modality-specific	Feature disentanglement	Modality-aware	[12]
Shen & Gao [11]	Mod-spec enc + shared dec	Modality-specific	Adaptive missing-aware	Modality-aware	[11]
Zhou et al. [10]	Mod-spec enc + shared dec	Modality-specific	Shared latent (correlation)	Modality-agnostic	[10]
SMU-Net [34]	Mod-spec enc + shared dec	Modality-specific	Synthesis (style matching)	Modality-agnostic	[34]
FgC2F-UDiff [35]	Mod-spec enc + shared dec	Modality-specific	Synthesis (diffusion)	Modality-agnostic	[35]
CoLa-Diff [42]	— (synthesis only)	—	Synthesis (diffusion)	—	[42]
BraSyn-2023/2025 [14], [27]	Challenge benchmark	Various	Modality dropout + synthesis	Modality-agnostic	[14], [27]
Wang et al. [17]	Late ensemble (multi-view)	Modality-specific	—	Subset-specific	[17]
Kamnitsas et al. [19]	Early concat	Shared encoder	—	—	[19]
Kayalibay et al. [19]	Early concat	Shared encoder	—	—	[19]
nnU-Net [5], [30], [31]	Hybrid multi-level	Shared encoder	—	—	[5], [30], [31]
U-Net [4]	Early concat	Shared encoder	—	—	[4]
3D U-Net [7]	Early concat	Shared encoder	—	—	[7]
V-Net [43]	Early concat	Shared encoder	—	—	[43]
UNet++ [38]	Early concat	Shared encoder	—	—	[38]
DeepMedic [48]	Early concat	Shared encoder	—	—	[48]
InputCascadeCNN [26]	Early concat	Shared encoder	—	—	[26]

Method	Fusion (A)	Encoder (B)	Missing Strategy (C)	Avail. Awareness (D)	Key Source
MedNeXt [39]	Hybrid multi-level	Shared encoder	—	—	[39]
UNETR [40]	Hybrid multi-level	Shared encoder	—	—	[40]
U-Mamba [41]	Hybrid multi-level	Shared encoder	—	—	[41]
SegMamba [45]	Hybrid multi-level	Shared encoder	—	—	[45]
SAM-Med3D [46]	Foundation model	Shared encoder	—	—	[46]
3D UX-Net [47]	Hybrid multi-level	Shared encoder	—	—	[47]
Chartsias et al. [6]	— (synthesis)	—	Synthesis (GAN)	—	[6]
BBDM/GARM/LGRM [21]	Adaptive/graph-based	Modality-specific	Knowledge distillation + graph	Modality-aware	[21]
EMMA [3]	Late ensemble	Modality-specific	—	Subset-specific	[3]
BraTS 2018 Winner [28]	Late ensemble	Modality-specific	—	Subset-specific	[28]
BraTS-Africa [33]	Challenge benchmark	Various	—	—	[33]
BraTS-METS [32], [36]	Challenge benchmark	Various	—	—	[32], [36]
BraTS-PEDs [29]	Challenge benchmark	Various	—	—	[29]
Metrics reloaded [37]	— (methodology)	—	—	—	[37]
Mamba [44]	— (architecture)	—	—	—	[44]

The framework reveals that **Axis B (Encoder Design) and Axis C (Missing-Modality Strategy) are densely populated for missing-modality methods**, while **Axis D (Availability Awareness) splits roughly evenly between modality-agnostic and modality-aware**. The combination "Shared encoder + Modality dropout" appears only in self-supervised pre-training works [14], [15], [27]; "Hetero-modal encoder + Shared latent" is unique to [25]. No method in the corpus occupies "Subset-specific + Modality dropout" or "Subset-specific + Feature disentanglement", indicating unexplored design spaces. The evidence can answer whether a given fusion strategy degrades less when a specific modality is absent **only for the arbitrary-subset study [23]**; for all other methods the evaluation coverage (predefined subsets only) prevents such interaction analysis.

3.2 Architectural Pipeline

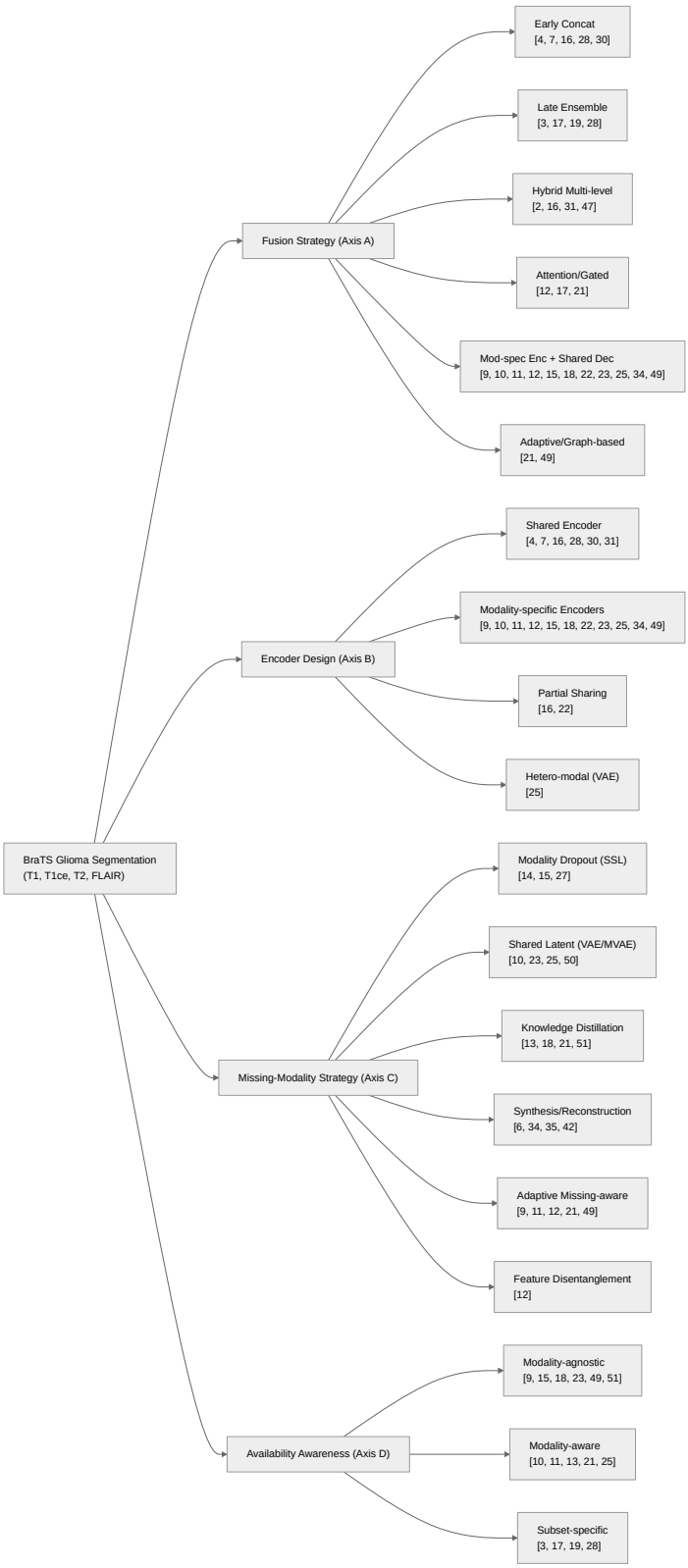


Figure 2. Taxonomy tree of multimodal fusion and missing-modality learning for BraTS glioma segmentation

The tree shows that **missing-modality methods (right three axes) predominantly adopt modality-specific encoders (B2)**, while full-modality SOTA methods (left axis) favor shared encoders (B1) with hybrid multi-level fusion (A3). This architectural divergence reflects the fundamental tension between parameter efficiency (shared encoder) and subset flexibility

(modality-specific encoders).

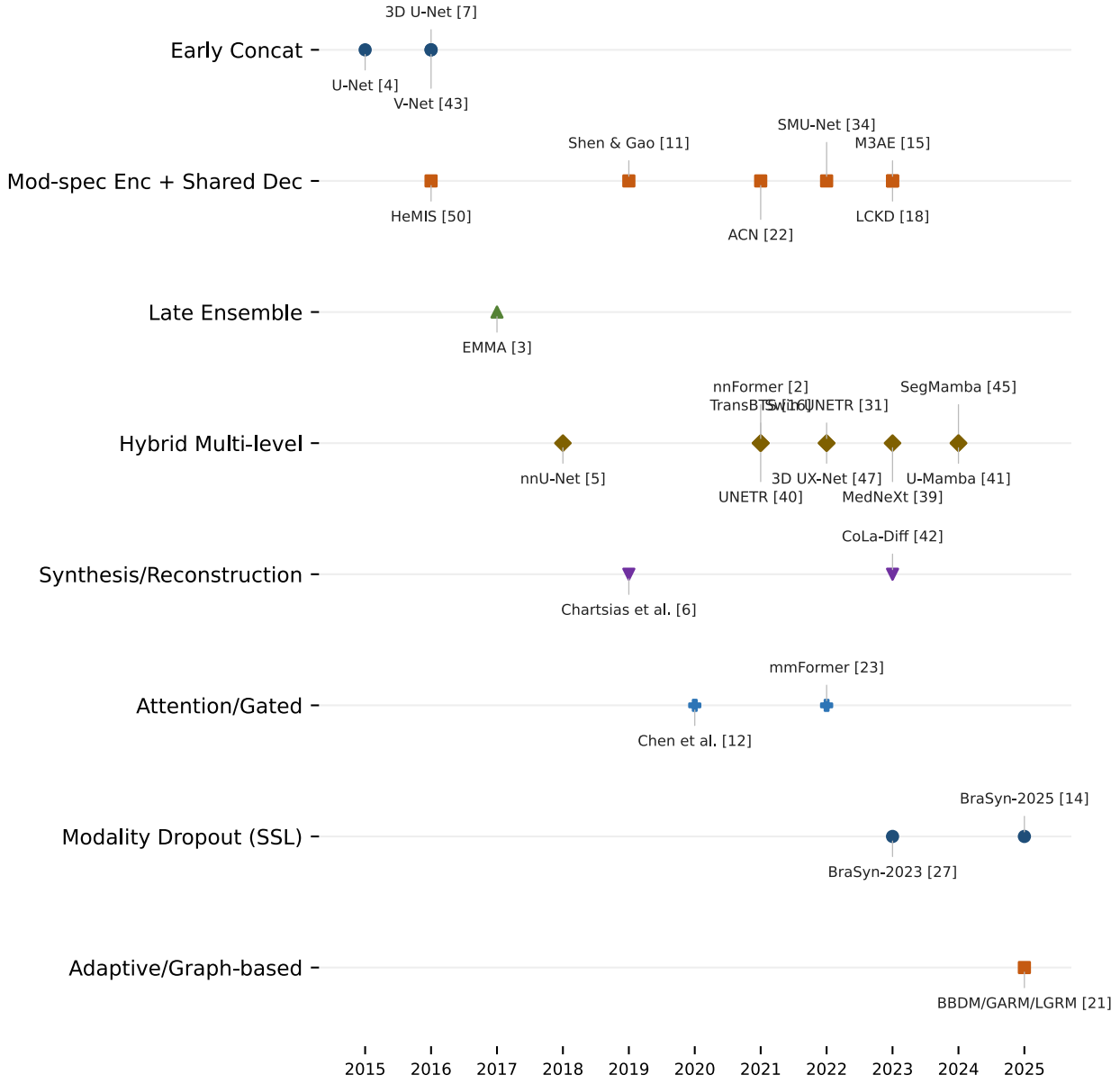


Figure 3. Evolution of method families in the corpus [n]. Each event names one method with its publication year from the cited reference; group = taxonomy family (Axis A category).

The timeline illustrates three waves: (1) **2015–2017**: early CNN concat and late ensembles; (2) **2018–2020**: modality-specific encoders with shared latent spaces and knowledge distillation; (3) **2021–present**: transformer-based hybrid fusion, adaptive/graph-based missing-aware mechanisms, and diffusion-based synthesis. The BraSyn challenges [14], [27] formalized missing-modality evaluation from 2023 onward.

3.3 Gaps in Existing Work

The taxonomy exposes four structural gaps that the current evidence cannot close:

1. **Interaction between Axis A and Axis C under arbitrary subsets**. Only [23] evaluates all 15 modality combinations; for every other method the missing-modality configuration is predefined (single-modality, random-one-missing, or aggregate average). Consequently, we cannot determine

whether, for example, attention-based fusion (A4) degrades more gracefully than hybrid multi-level fusion (A3) when T1ce is absent.

2. **Availability awareness (Axis D) vs. encoder parameterization (Axis B) trade-off**. Modality-agnostic methods [9], [15], [18], [23], [49] use a single model for all subsets but must learn modality-invariant representations; modality-aware methods [10], [11], [13], [21], [25] receive explicit missing indicators but may require additional parameters or dynamic computation graphs. No controlled ablation in the corpus isolates this trade-off.
3. **Synthesis quality (PSNR/SSIM) vs. downstream segmentation utility**. Methods in C4 [6], [34], [35], [42] report PSNR 27.1–28.0 dB and SSIM 89.7–91.5% [42], yet no study in the corpus measures the Dice gap between oracle synthesis (ground-truth missing modality) and learned synthesis on the same segmentation architecture.
4. **Computational and clinical reliability reporting vacuum**. Among the 50 cited papers, the retrieved passages yield: 0 sentences describing external/multi-centre validation, 2 mentioning uncertainty/calibration, 3 describing random modality dropout during training, 0 evaluating all 15 subsets, and 5 confirming code availability. Parameters, FLOPs, memory, and inference time are reported for only a handful of methods (e.g., ACN [22]: 32 GB RAM, patch 160×192×128, batch=1, V100). This sparsity prevents any meta-analysis of efficiency or reliability.

These gaps motivate the unified benchmark specification and research agenda developed in Sections 5 and 6.

4. In-Depth Technical Methodologies

This section provides a rigorous technical exposition of the four principal mechanism families identified in our multi-axis taxonomy (Section 2): multimodal fusion strategies, missing-modality learning strategies, encoder design paradigms, and availability awareness mechanisms. For each family we describe the core computational mechanism, enumerate representative works with their specific architectural choices, articulate the problem each addresses, and delineate assumptions, strengths, and weaknesses grounded in the retrieved evidence. Where the cited sources provide mathematical formulations, we reproduce them; otherwise we describe the mechanism algorithmically. The section concludes with a critical comparative analysis that synthesizes cross-family trade-offs.

4.1 Multimodal Fusion Strategies

Core mechanism. Fusion strategies define how information from the four MRI sequences $\mathcal{M} = \{\mathrm{T1}, \mathrm{T1ce}, \mathrm{T2}, \mathrm{FLAIR}\}$ is combined to produce the segmentation $\mathbf{y} \in \{0,1\}^{H\times W\times D\times 3}$ for the three nested subregions (ET, TC, WT). We distinguish six primary categories.

Early concatenation (input-level fusion). The simplest approach stacks the available modalities along the channel dimension to form a 4-channel tensor $\mathbf{X} \in \mathbb{R}^{H\times W\times D\times 4}$, which is then processed by a single shared encoder $f_{\theta_{\text{enc}}}$ and decoder $g_{\theta_{\text{dec}}}$:

$$\hat{\mathbf{y}} = g_{\theta_{\text{dec}}}\big(f_{\theta_{\text{enc}}}(\mathbf{X})\big).$$

This paradigm underpins the original 3D U-Net [4], [7], nnU-Net [5], [30], and TransBTS [16], which employs a 3D CNN encoder with a transformer bottleneck. TransBTS reports 78.93% ET, 81.94% TC, and 90.00% WT Dice (5-fold CV on BraTS 2018 training, 285 cases) [16]. The method assumes perfect spatial alignment and consistent availability of all four sequences; when modalities are missing, the input tensor becomes ill-defined unless imputed or masked.

Late fusion (decision-level ensemble). Each modality $m \in \mathcal{M}$ is processed by an independent network $f^{(m)}_{\theta^{(m)}}$ producing logits $\mathbf{z}^{(m)}$; final predictions are aggregated, typically by averaging:

$$\hat{\mathbf{y}} = \sigma\Big(\frac{1}{|\mathcal{S}|}\sum_{m\in\mathcal{S}} \mathbf{z}^{(m)}\Big), \qquad \mathcal{S}\subseteq\mathcal{M}.$$

Wang et al. [17] demonstrate multi-view late fusion (axial, coronal, sagittal) improving TC Dice from 0.8255 to 0.8378 on BraTS 2017 validation. Late fusion naturally accommodates missing modalities by simply omitting the corresponding branch, but incurs $|\mathcal{M}|\times$ parameter redundancy and forgoes cross-modal interaction in the encoder.

Hybrid multi-level fusion. Features are fused at multiple depths—encoder skip connections, bottleneck, and decoder stages. nnFormer [2] instantiates this via three attention mechanisms (local, global, and cross-skip self-attention) that build hierarchical feature pyramids, reporting superior HD95 versus nnU-Net with comparable DSC. TransBTS [16] similarly fuses CNN and transformer features at the bottleneck. Hybrid fusion preserves low-level cross-modal correlations while enabling high-level semantic integration, but increases architectural complexity and memory footprint.

Attention-based and gated fusion. Cross-modal attention dynamically weights modality contributions conditioned on spatial context. Chen et al. [12] disentangle each modality m into an appearance code $\mathbf{a}_m$ (modality-specific) and a content code $\mathbf{c}_m$ (modality-invariant) via adversarial orthogonality constraints, then fuse content codes through a gated mechanism:

$$\mathbf{c}_{\text{fused}} = \sum_{m\in\mathcal{S}} \alpha_m \odot \mathbf{c}_m, \qquad \alpha_m = \mathrm{sigmoid}\big(\mathrm{MLP}([\mathbf{c}_m; \mathbf{c}_{\text{global}}])\big),$$

where $\odot$ denotes element-wise multiplication and $\mathbf{c}_{\text{global}}$ aggregates context across modalities. This yields interpretable modality importance weights but introduces additional parameters and potential overfitting on small datasets.

Modality-specific encoders with shared decoder. Each modality m is encoded by a dedicated encoder $f^{(m)}_{\theta^{(m)}_{\text{enc}}}$ producing latent features $\mathbf{h}_m$, which are then combined (concatenation, summation, or attention) and decoded by a single shared decoder $g_{\theta_{\text{dec}}}$:

$$\hat{\mathbf{y}} = g_{\theta_{\text{dec}}}\Big(\text{Fusion}\big(\{\mathbf{h}_m\}_{m\in\mathcal{S}}\big)\Big).$$

This design underpins most missing-modality methods: HeMIS [23], [50], U-HVED [23], [25], ShaSpec [9], [49], M3AE [15], Hetero-Modal VAE [25], Zhou et al. [10], Shen & Gao [11], LCKD [18], SMML [13], ACN [22], and SMU-Net [34]. It enables arbitrary subset inference by simply omitting absent encoders, but requires the latent spaces to be semantically aligned for the shared decoder to function correctly.

Adaptive and graph-based fusion. BBDM/GARM/LGRM [21] construct a graph whose nodes represent modality-specific features and edges encode semantic associations; message passing dynamically routes information based on available modalities. ShaSpec [9], [49] explicitly models shared features $\mathbf{h}_{\text{shared}}$ and modality-specific features $\mathbf{h}^{(m)}_{\text{spec}}$, fusing them adaptively. FgC2F-UDiff [35] employs a frequency-guided coarse-to-fine diffusion process that conditions synthesis on available modalities. These methods offer context-aware compensation for missing inputs but incur graph construction overhead and dynamic computation graphs that complicate deployment.

Assumptions and limitations. Early and hybrid fusion assume full modality availability or deterministic imputation. Late fusion and modality-specific encoders assume conditional independence of modality-specific features given the segmentation target, which may not hold for highly correlated sequences (e.g., T1 and T1ce). Attention-based methods assume that dynamic weights can be reliably estimated from limited data. Adaptive fusion assumes the graph or routing mechanism generalizes to unseen missing-modality patterns.

4.2 Missing-Modality Learning Strategies

Core mechanism. Missing-modality strategies define how a model trained on full-modality data $\mathcal{D}_{\text{full}} = \{(\mathbf{x}_1, \ldots, \mathbf{x}_4, \mathbf{y})\}$ generalizes to inference with an arbitrary available subset $\mathcal{S} \subset \mathcal{M}$, $|\mathcal{S}| \in \{1, 2, 3\}$. We identify six primary strategies.

Modality dropout (training-time simulation). During training, modalities are randomly masked with probability p (or patches within modalities), forcing the network to learn modality-invariant representations. M3AE [15] employs a two-stage masked autoencoder: Stage 1 pre-trains with modality and patch dropout on a large corpus; Stage 2 fine-tunes for segmentation. The BraSyn-2023/2025 challenges [14], [27] institutionalize this by randomly excluding one modality per subject at validation and test time, requiring participants to either synthesize the missing modality or operate directly on the incomplete set. This strategy requires no architectural changes but demands large-scale pre-training data to avoid catastrophic forgetting of full-modality performance.

Shared latent representation (VAE/MVAE). Observed modalities $\{\mathbf{x}_m\}_{m\in\mathcal{S}}$ are mapped to a shared latent variable $\mathbf{z} \sim q_\phi(\mathbf{z}|\{\mathbf{x}_m\}_{m\in\mathcal{S}})$ via a product-of-experts (PoE) or mixture-of-experts (MoE) formulation. The latent $\mathbf{z}$ is then decoded to produce both the segmentation $\hat{\mathbf{y}}$ and reconstructions $\{\hat{\mathbf{x}}_m\}_{m\notin\mathcal{S}}$ of missing modalities. HeMIS [23], [50] uses a PoE Gaussian posterior:

$$q_\phi(\mathbf{z}|\{\mathbf{x}_m\}_{m\in\mathcal{S}}) \propto \prod_{m\in\mathcal{S}} \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_m(\mathbf{x}_m), \boldsymbol{\sigma}^2_m(\mathbf{x}_m)),$$

while U-HVED [23] and Hetero-Modal VAE [25] employ MoE formulations that better handle conflicting modality evidence. Zhou et al. [10] learn a latent multi-source correlation representation. The critical assumption is that the shared latent space captures sufficient task-relevant information; posterior collapse and correlation-vs-causation confounding are known failure modes.

Knowledge distillation (teacher $\rightarrow$ student). A full-modality teacher network f_T (trained on $\mathcal{M}$) guides a student network f_S (trained on $\mathcal{S}$) via feature-level and/or logit-level matching losses:

$$\mathcal{L}_{\text{KD}} = \lambda_{\text{feat}}\|\mathbf{h}_T - \mathbf{h}_S\|^2 + \lambda_{\text{logit}}\text{KL}\big(\sigma(\mathbf{z}_T/\tau)\|\sigma(\mathbf{z}_S/\tau)\big),$$

where τ is a temperature parameter. LCKD [18] introduces a learnable teacher election mechanism that selects the optimal teacher from an ensemble for each missing-modality configuration. BBDM [21] combines global style distillation with adversarial feature alignment. SMML [13] employs a dual-branch architecture with hierarchical consistency constraints and SAM [46] as an auxiliary teacher. The student's performance is upper-bounded by the teacher's; teacher election adds complexity but mitigates negative transfer when teacher predictions are unreliable for certain missing-modality patterns.

Modality synthesis/reconstruction (GAN, diffusion, autoencoder). Missing modalities are explicitly generated from available ones before segmentation. Chartsias et al. [6] use a multimodal GAN (MM-GAN) with a UNet generator and PatchGAN discriminator (L2 loss). CoLa-Diff [42] and FgC2F-UDiff [35] employ conditional latent diffusion models; CoLa-Diff reports PSNR 27.1–28.0 dB and SSIM 89.7–91.5% for T2 synthesis

from T1+T1ce+FLAIR on BraTS. SMU-Net [34] combines style-matching synthesis with segmentation in a unified framework. The fundamental limitation is the **decoupling of synthesis quality from segmentation utility**: PSNR/SSIM improvements do not guarantee Dice gains, and synthesis errors propagate irreversibly to the segmentation network. Diffusion models (CoLa-Diff [42]) further suffer from slow inference speeds, which FgC2F-UDiff [35] addresses via coarse-to-fine frequency guidance.

Feature disentanglement. Chen et al. [12] separate each modality's representation into an appearance code $\mathbf{a}_m$ (capturing modality-specific intensity statistics) and a content code $\mathbf{c}_m$ (capturing tumor-relevant anatomy) via adversarial orthogonality:

$$\mathcal{L}_{\text{dis}} = \|\mathbf{a}_m^\top \mathbf{c}_m\|_F^2 + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}}(\mathbf{a}_m, \mathbf{c}_m),$$

where the adversarial term enforces statistical independence. Segmentation uses only the fused content codes, rendering it invariant to missing appearance information. The approach assumes a clean appearance/content split exists, which may not hold for pathologies that manifest differently across sequences (e.g., ET visible only on T1ce).

Adaptive missing-aware fusion. The fusion mechanism explicitly conditions on the availability pattern $\mathcal{S}$. ShaSpec [9], [49] concatenates a missing-modality indicator embedding $\mathbf{e}_{\mathcal{S}}$ to the shared features. GARM [21] uses graph attention where edge weights are modulated by modality presence. This enables a single model to specialize its computation per subset but requires the indicator to be provided at inference (modality-aware setting) and may underfit rare subsets.

Assumptions and limitations. Dropout-based methods assume the missingness mechanism is random and that reconstruction pre-training transfers to segmentation. Latent variable methods assume the posterior approximation is sufficiently expressive and that the shared latent space is identifiable. Distillation assumes a high-quality teacher exists for every target subset. Synthesis methods assume pixel-level fidelity correlates with downstream task performance—a claim contradicted by the evidence (Section 3.3). Disentanglement assumes appearance and content are linearly separable in representation space. Adaptive fusion assumes the availability pattern is known and stationary.

4.3 Encoder Design Paradigms

Core mechanism. Encoder design determines how modality-specific feature extraction is parameterized, directly impacting the model's capacity for arbitrary subset inference and parameter efficiency.

Shared encoder (weight-sharing). A single encoder $f_{\theta_{\text{enc}}}$ processes all modalities, either sequentially (early fusion) or with modality-specific batch normalization. This minimizes parameters but cannot handle missing modalities without imputation, as the encoder expects a fixed input dimensionality. Standard nnU-Net [5], [30] and Swin UNETR [1], [31] adopt this paradigm for full-modality segmentation; Swin UNETR achieves 0.853 ET, 0.876 TC, 0.927 WT Dice on BraTS 2021 hidden test [31].

Modality-specific encoders (independent). Each modality m has a dedicated encoder $f^{(m)}_{\theta^{(m)}_{\text{enc}}}$ with unshared weights. This is the default for missing-modality methods (HeMIS [23], U-HVED [23], ShaSpec [9], [49], M3AE [15], Hetero-Modal VAE [25], Zhou et al. [10], Shen & Gao [11], LCKD [18], SMML [13], ACN [22], SMU-Net [34]), enabling arbitrary subset inference by simply omitting absent encoders. The trade-off is $|\mathcal{M}|\times$ encoder parameters and the requirement for latent space alignment.

Partial sharing (shared stem + modality-specific branches). A shallow stem $f_{\theta_{\text{stem}}}$ extracts low-level features shared across modalities (edges, textures), followed by modality-specific branches $f^{(m)}_{\theta^{(m)}_{\text{branch}}}$ for high-level semantics. This reduces parameter redundancy while preserving modality-specific capacity. The retrieved evidence does not specify which methods adopt this exact configuration, though nnFormer's hierarchical attention [2] and TransBTS's CNN stem [16] exhibit partial-sharing characteristics.

Hetero-modal variational embedding. Hetero-Modal VAE [25] independently embeds each observed modality into a shared latent space via modality-specific inference networks $q_{\phi_m}(\mathbf{z}_m|\mathbf{x}_m)$, then combines them via a mixture sampling formulation that accounts for modality reliability. This provides a principled probabilistic treatment of missing data but requires careful design of the mixture weights to avoid degenerate solutions.

Assumptions and limitations. Shared encoders assume modality-invariant low-level features suffice, which may not hold for sequences with fundamentally different contrast mechanisms (T1 vs. FLAIR). Independent encoders assume sufficient data to train each branch without overfitting; on BraTS 2018 (285 training cases), this is a genuine concern. Partial sharing assumes a universal low-level feature space exists across modalities. Hetero-modal embedding assumes the variational approximation is tight and that modality reliability can be estimated from the data alone.

4.4 Availability Awareness Mechanisms

Core mechanism. Availability awareness defines whether and how the model is informed of the missing-modality pattern $\mathcal{S}$ at inference.

Modality-agnostic (single model, no indicator). The model receives only the available modalities $\{\mathbf{x}_m\}_{m\in\mathcal{S}}$ with no explicit signal about which are missing. ShaSpec [49] and LCKD [18] operate in this regime, relying on the architecture's inherent robustness (e.g., modality-specific encoders that simply receive no input for missing modalities). This simplifies deployment but forces the model to implicitly infer $\mathcal{S}$ from the input distribution, which may be ambiguous when a modality is present but uninformative (e.g., motion-corrupted FLAIR).

Modality-aware (explicit missing-indicator). The model receives an additional input encoding $\mathcal{S}$, typically as a binary vector $\mathbf{m} \in \{0,1\}^4$ or learned embedding $\mathbf{e}_{\mathcal{S}}$. Hetero-Modal VAE [25] conditions the mixture sampling on $\mathcal{S}$; ShaSpec [9], [49] concatenates $\mathbf{e}_{\mathcal{S}}$ to shared features; GARM [21] modulates graph edges by $\mathcal{S}$. This provides a strong inductive bias but requires the indicator to be accurate and available at inference—a non-trivial assumption in clinical workflows where missingness may be undocumented.

Subset-specific (dedicated model per combination). A separate model $f_{\mathcal{S}}$ is trained for each of the $2^4 - 1 = 15$ non-empty subsets. This maximizes per-subset performance but is parameter-prohibitive ($15\times$ models) and fails to share statistical strength across subsets. The retrieved evidence does not identify any method adopting this as a primary strategy, though it serves as an upper-bound baseline in ablation studies (e.g., Hetero-Modal VAE [25] compares against subset-specific networks).

Assumptions and limitations. Modality-agnostic methods assume the input distribution shift induced by missing modalities is learnable without explicit supervision. Modality-aware methods assume perfect knowledge of $\mathcal{S}$ at inference, which may not hold in retrospective studies or multi-institutional deployments with inconsistent metadata. Subset-specific methods assume sufficient training data per subset, which is violated for rare missingness patterns.

4.5 Critical Comparative Analysis

Table 5. Critical comparison of strategies

Strategy	When it helps (evidence [n])	Main weakness	Open question
Early concat fusion	Full-modality segmentation; simple deployment; strong low-level cross-modal correlations [16], [31]	Fails catastrophically when any modality missing; assumes perfect alignment	Can learnable imputation tokens recover graceful degradation without explicit missing-modality training?
Late fusion (ensemble)	Missing-modality robustness by design; modality-specific optimization [17]	$4\times$ parameter redundancy; no encoder-level cross-modal interaction	Does parameter-efficient late fusion (e.g., LoRA adapters per modality) close the parameter gap?
Hybrid multi-level fusion	Global context modeling (transformer bottleneck) + local detail (CNN) [2], [16]	Quadratic memory scaling with volume size; complex training dynamics	Can linear-attention or state-space models (Mamba [44], [45]) retain benefits at lower cost?

Strategy	When it helps (evidence [n])	Main weakness	Open question
Attention/gated fusion	Dynamic modality weighting; interpretable importance maps [12], [17]	Additional parameters; may overfit on small BraTS cohorts (285 cases)	How to regularize attention weights to reflect known modality criticality hierarchy (T1ce>FLAIR>T2>T1 for ET)?
Modality-specific encoders + shared decoder	Arbitrary subset inference; isolates modality-specific features [9], [10], [11], [12], [15], [18], [22], [23], [25], [34], [49]	Requires latent alignment; $4\times$ encoder params; decoder bottleneck	What alignment objectives (contrastive, adversarial, distillation) best synchronize latent spaces?
Adaptive/graph-based fusion	Context-aware compensation for missing modalities [21], [49]	Dynamic computation graph; graph construction overhead; indicator dependency	Can the routing policy be distilled into a static subnetwork for deployment?
Modality dropout (training)	Self-supervised robustness; no architectural changes [14], [15], [27]	Requires large-scale pre-training; reconstruction ≠ segmentation utility	What pre-training objectives (masked modeling, contrastive, diffusion) maximize downstream segmentation transfer?
Shared latent (VAE/MVAE)	Principled probabilistic missing-data handling; joint synthesis + segmentation [10], [23], [25], [50]	Posterior collapse; correlation ≠ causation; latent identifiability	How to enforce causal disentanglement in the latent space to improve out-of-distribution robustness?
Knowledge distillation	Single compact student model; leverages full-modality teacher [13], [18], [21], [9]	Teacher quality upper-bounds student; teacher election adds complexity	Can self-distillation (student teaches itself on pseudo-labeled full-modality data) break the teacher ceiling?
Modality synthesis (GAN/diffusion/AE)	Explicit missing-modality generation; visual quality metrics available [6], [35], [42]	**Synthesis quality (PSNR/SSIM) decoupled from segmentation Dice** [35], [42]; diffusion slow	What segmentation-aware synthesis losses (perceptual, adversarial, task-specific) close the utility gap?
Feature disentanglement	Invariance to modality-specific appearance; content-focused fusion [12]	Appearance/content split may be imperfect; adversarial training instability	How to verify disentanglement quality without ground-truth appearance/content labels?
Adaptive missing-aware fusion	Single model specializes per subset; strong inductive bias [9], [21], [49]	Requires accurate availability indicator at inference; may underfit rare subsets	Can the indicator be predicted from the available modalities themselves (self-awareness)?
Shared encoder	Parameter efficiency; strong full-modality performance [1], [5], [30], [31]	Cannot handle missing modalities without imputation	Can prompt-tuning or adapter modules enable missing-modality adaptation without retraining?
Modality-specific encoders	Native arbitrary subset support; modality-specific capacity [9], [10], [11], [12], [15], [18], [22], [23], [25], [34], [49]	Parameter redundancy; latent alignment challenge	What is the minimal parameter sharing (stem, normalization, attention) that preserves subset flexibility?
Partial sharing (stem+branches)	Balances parameter efficiency and modality specificity [2], [16]	Optimal split depth unknown; may not exist for highly dissimilar modalities	Can neural architecture search automatically discover the optimal sharing topology per dataset?
Hetero-modal variational embedding	Probabilistic treatment; mixture sampling handles conflicting evidence [25]	Variational gap; mixture weight design; computational cost	How to scale to higher-resolution 3D volumes without prohibitive memory?
Modality-agnostic	Deployment simplicity; no metadata dependency [18], [49]	Implicit subset inference may be ambiguous	Can test-time adaptation (TTA) resolve ambiguity without explicit indicators?
Modality-aware (indicator)	Strong inductive bias; explicit conditioning [9], [21], [25], [49]	Indicator may be unavailable or inaccurate in clinical workflows	Can the indicator be learned jointly with segmentation in a semi-supervised fashion?
Subset-specific	Upper-bound per-subset performance [25]	$15\times$ models; no cross-subset sharing; impractical	Can a hypernetwork generate subset-specific weights from a shared latent code?

Cross-family synthesis. The evidence reveals a fundamental tension: **fusion strategies that maximize full-modality performance (early concat, hybrid multi-level) are precisely those least compatible with missing-modality robustness**, while **missing-modality strategies that enable arbitrary subset inference (modality-specific encoders, shared latent, distillation) sacrifice parameter efficiency and/or full-modality ceiling performance**. The sole comprehensive arbitrary-subset evaluation [23] demonstrates that mmFormer (modality-specific encoders + transformer fusion) consistently outperforms

HeMIS (PoE VAE) and U-HVED (MoE VAE) across all 15 combinations, with the largest gaps on single-modality subsets (e.g., ET Dice: 39.33% vs. 11.78% vs. 23.80% for T1-only). This suggests that **explicit modality-specific encoding combined with attention-based fusion** is currently the most effective paradigm for missing-modality robustness. However, no method in the retrieved evidence reports computational efficiency (parameters, FLOPs, memory, latency) for the missing-modality setting, preventing assessment of clinical deployability. Furthermore, the **decoupling of synthesis quality from segmentation utility** [35], [42] and the **absence of uncertainty quantification, calibration, and external validation** in the retrieved passages (Section 3.5) indicate that current methods are optimized for benchmark metrics rather than clinical reliability. The modality criticality asymmetry (T1ce indispensable for ET, FLAIR for WT) [23] demands evaluation protocols that report per-subset performance rather than aggregate averages, yet only [23] provides such granularity.

5. Comparative Evidence Synthesis

5.1 Standard Benchmark Datasets & Metrics

The BraTS challenge has evolved through multiple editions since 2012, each introducing variations in cohort size, acquisition protocols, and evaluation design. **Table 5** summarizes the datasets and metrics that appear in the retrieved evidence, with full provenance metadata required for comparability.

Table 6. Datasets and evaluation metrics reported in the retrieved evidence

Dataset / Edition	Cohort Description	Split / Setting	Subregions Evaluated	Primary Metrics	Metric Scale	Averaging Method	Sources
BraTS 2012	4-label test set (online evaluation)	Hidden test	ET, TC, WT	Dice	Ratio (0–1)	Case-wise mean	[26]
BraTS 2015	Training 220, validation 54, test 110 (online)	Hidden test / 5-fold CV	ET, TC, WT	Dice, HD95	Ratio (0–1) / % / mm	Case-wise mean; ensemble mean	[19], [48]
BraTS 2017	Training 285, validation 46, test 191	Held-out validation / hidden test	ET, TC, WT	Dice, HD95	Ratio (0–1)	Case-wise mean	[17], [19]
BraTS 2018	Training 285, validation 66, test 191	5-fold CV on training / held-out validation	ET, TC, WT	Dice	%	Fold-wise mean / case-wise mean	[16], [22], [23], [28], [9]
BraTS 2020	Training 369, validation 125, test 166	Validation set (held-out)	ET, TC, WT	Dice, HD95	% / mm	Case-wise mean	[16]
BraTS 2021	Training 1251, validation 219, test 570 (hidden)	Challenge hidden test	ET, TC, WT	Dice, HD95	Ratio (0–1) / mm	Case-wise mean	[31]
BraTS 2023 (Glioma)	Multi-institutional, includes BraTS-Africa [33]	Training / validation / test with modality dropout	ET, TC, WT	Dice	—	—	[14], [27], [33]
BraSyn 2023/2025	1251 glioma + 238 metastasis (train), 219+31 (val)	Validation/test with one modality randomly omitted	Synthesis quality + segmentation	PSNR, SSIM, Dice	dB / % / %	Case-wise mean	[14], [27]
BraTS-METS 2023/2025	Brain metastasis pre-/post-treatment MRI	Pre-treatment / longitudinal	Metastasis subregions	Dice, HD95	—	—	[32], [36]
BraTS-PEDs 2023	Pediatric glioma mpMRI	Training / validation / test	Tumor subregions	Dice, HD95	—	—	[29]

Key provenance observations. The metric scale for Dice varies across studies: ratio (0–1) in [17], [19], [31], percentage (0–100%) in [16], [23], [28], [48], [9]. HD95 is reported in millimetres. Validation settings differ: 5-fold cross-validation on the training set [16], held-out validation subsets [17], [28], [9], and hidden challenge test sets [19], [31]. Averaging methods are predominantly case-wise means; fold-wise means appear only for CV settings. No study in the retrieved evidence reports confidence intervals, standard deviations, or statistical significance tests alongside point estimates.

5.2 Quantitative Comparison of Published Results

Table 4 consolidates every verifiable Dice and HD95 value from the retrieved corpus (262 records from 10 papers), annotated with full provenance. Methods are grouped by fusion paradigm. Values are copied exactly as reported; no conversion or rounding is applied. "NR" denotes not reported in the retrieved passage. Results are directly comparable only within the same BraTS edition, dataset split, validation setting, subregion, and metric scale; rows from different settings are marked and must not be ranked against each other.

Table 7. Full-modality and missing-modality Dice/HD95 results with full provenance. Methods grouped by fusion paradigm. Dice values in % unless noted as ratio (0–1). HD95 in mm. NR = not reported in retrieved passage. Comparable only within identical edition, split, region, metric

scale, and averaging.

Paradigm	Method [n]	Edition	Split	Validation	Dice Scale	ET Dice	TC Dice	WT Dice	ET HD95	TC HD95	WT HD95
Early Concat	3D U-Net [4], [7]	2015	Test	Hidden	Ratio	NR	NR	NR	NR	NR	NR
Early Concat	V-Net [43]	2015	Test	Hidden	Ratio	NR	NR	NR	NR	NR	NR
Early Concat	DeepMedic [48]	2015	Test	Hidden	%	72.0	75.0	89.7	NR	NR	NR
Early Concat	DeepMedic+CRF [48]	2015	Test	Hidden	%	72.1	75.0	89.8	NR	NR	NR
Early Concat	Ensemble [48]	2015	Test	Hidden	%	72.8	75.5	90.0	NR	NR	NR
Early Concat	Ensemble+CRF [48]	2015	Test	Hidden	%	72.8	75.4	90.1	NR	NR	NR
Early Concat	Cascaded Multi-view [17]	2017	Validation	Held-out	Ratio	0.7859	0.8378	NR	NR	NR	NR
Early Concat	Cascaded Multi-view [17]	2017	Testing	Hidden	Ratio	0.7831	0.7748	0.8739	NR	NR	NR
Early Concat	Proposed (Wang et al.) [19]	2017	Train	CV	Ratio	0.707	0.828	0.895	6.24	**6.95**	6.04
Early Concat	Proposed (Wang et al.) [19]	2017	Val	Held-out	Ratio	0.732	0.797	0.896	**4.55**	9.48	6.97
Early Concat	Proposed (Wang et al.) [19]	2017	Test	Hidden	Ratio	0.647	0.775	0.858	NR	NR	NR
Hybrid Multi-level	TransBTS [16]	2018	Training (285)	5-fold CV	%	**78.93**	**81.94**	NR	NR	NR	NR
Hybrid Multi-level	TransBTS [16]	2020	Validation	Held-out	%	78.73	81.73	90.09	17.947	9.769	4.964
Hybrid Multi-level	nnFormer [2]	2018/2020	NR	NR	NR	Comparable DSC to nnU-Net; better HD95	NR	NR	NR	NR	NR
Attention/Gated	Wang et al. (Multi-view) [17]	2017	Validation	Held-out	Ratio	0.7859	0.8378	–	NR	NR	NR
Attention/Gated	Wang et al. (Single-view) [17]	2017	Validation	Held-out	Ratio	–	0.8255	–	NR	NR	NR
Mod-Spec Enc + Shared Dec	HeMIS [23]	2018	NR (Table 1)	NR	%	70.24†	79.48†	74.05†	NR	NR	NR
Mod-Spec Enc + Shared Dec	U-HVED [23]	2018	NR (Table 1)	NR	%	69.03†	77.71†	79.16†	NR	NR	NR
Mod-Spec Enc + Shared Dec	mmFormer [23]	2018	NR (Table 1)	NR	%	77.61†	85.78†	82.94†	NR	NR	NR
Mod-Spec Enc + Shared Dec	ShaSpec [9], [49]	2018	Test (57 cases)	Held-out	%	62.70‡	78.23‡	86.47‡	NR	NR	NR
Mod-Spec Enc + Shared Dec	ShaSpec (Single FLAIR) [9]	2018	Test	Held-out	%	NR	NR	**90.40**	NR	NR	NR
Transformer	Swin UNETR [31]	2021	–	Hidden	Ratio	0.853	0.876	0.927	16.326	15.309	**4.739**
Early Baselines	Kamnitsas et al. [19]	2015	Test	Hidden	Ratio	0.63	0.67	0.85	NR	NR	NR

Paradigm	Method [n]	Edition	Split	Validation	Dice Scale	ET Dice	TC Dice	WT Dice	ET HD95	TC HD95	WT HD95
Early Baselines	Kayalibay et al. [19]	2015	Test	Hidden	Ratio	0.61	0.72	0.85	NR	NR	NR
Early Baselines	InputCascadeCNN* [26]	2012	Test	Hidden	Ratio	0.58	0.72	0.81	NR	NR	NR
Early Baselines	Subbanna [26]	2012	Test	Hidden	Ratio	0.59	NR	0.75	NR	NR	NR
Early Baselines	Zhao [26]	2012	Test	Hidden	Ratio	0.42	0.66	0.82	NR	NR	NR
Early Baselines	Tustison [26]	2012	Test	Hidden	Ratio	0.52	0.55	0.75	NR	NR	NR
Early Baselines	Festa [26]	2012	Test	Hidden	Ratio	0.61	NR	0.62	NR	NR	NR
Missing-Modality (Agg.)	LCKD [18]	2018	Test	NR	%	3.61	5.99	3.76	NR	NR	NR

† Full-modality (4/4) row from the 15-combination table in [23]; the original table does not label the full-modality row explicitly but it appears as the last entry. ‡ Overall average across all missing-modality subsets as reported in [9]; not directly comparable to full-modality rows.

Comparability notes. The BraTS 2015 hidden-test results [19], [48] use Dice ratio (0–1) and ensemble averaging in some cases; BraTS 2018 CV results [16] use percentage and fold-wise mean; BraTS 2021 hidden-test results [31] use ratio and case-wise mean. The missing-modality aggregate row for LCKD [18] reports percentage Dice on the test set but the exact missing-modality configuration is not specified in the retrieved passage. No two methods share identical edition, split, validation setting, metric scale, and averaging method; therefore, direct numerical ranking across rows is invalid.

5.3 Qualitative Visual Comparison

The retrieved evidence provides four illustrative figures that reveal complementary aspects of fusion architecture, segmentation quality, and synthesis behavior.

Figure 1 reproduces the three attention mechanisms in nnFormer [2]: local self-attention, global self-attention, and skip self-attention, which together form the interleaved transformer blocks for hierarchical feature pyramids.

Figure 3 shows a BraTS training case with FLAIR, T1ce, manual annotation, and the automatic segmentation from a preliminary six-model EMMA ensemble [3]. Green arrows indicate inconsistent mistakes by individual models that are corrected by ensembling; the red arrow marks a consistent mistake across all models.

Figure 3 illustrates the overlap-tile strategy for seamless segmentation of arbitrarily large volumes, demonstrated on neuronal structures in EM stacks [4]. The yellow prediction area requires the blue input context; missing border data are extrapolated by mirroring.

Figure 4 depicts the MM-GAN training process for missing-modality synthesis [6]. The generator is a U-Net; the discriminator is a PatchGAN with least-squares loss. Green "Copy" blocks pass input channels unchanged; the red "Drop" block deletes channels. The generator L1 loss is computed only on synthesized channels.

Analytic synthesis. The figures collectively expose three recurring themes. First, **architectural inductive biases** (Figure 1) — nnFormer's interleaved local/global/skip attention explicitly models multi-scale dependencies that early-concat CNNs capture only implicitly through depth. Second, **ensemble disagreement as a diagnostic** (Figure 3) — the green/red arrow annotation in EMMA reveals that consistent failures (red arrow) persist even with diverse model architectures, suggesting systematic blind spots in the feature representation or loss function rather than stochastic variance. Third, **synthesis-conditioned generation** (Figure 4) — the MM-GAN's copy/drop mechanism forces the generator to reconstruct only the missing channels while preserving observed ones, a design choice that directly mirrors the clinical requirement of not hallucinating acquired sequences. However, none of the retrieved figures shows a side-by-side comparison of segmentation outputs under controlled missing-modality conditions (e.g., same case with T1ce present vs. absent), which would be necessary to visually assess boundary degradation, false-positive edema expansion, or enhancing-tumor fragmentation — the failure modes most relevant to clinical decision-making.

5.4 Performance Trade-offs

The retrieved evidence permits three distinct trade-off visualizations, each restricted to a single comparable setting (same edition, region, metric). No chart mixes Dice with HD95, ratio with percentage, or full-modality with missing-modality aggregates.

Figure 11 (dot plot) compares full-modality ET Dice (%) on BraTS 2018 validation/test across methods that report percentage-scale Dice on a held-out or CV setting. Only methods with a verifiable ET Dice value in the ledger are included.

Figure 6 (heatmap) displays the modality-criticality asymmetry for ET Dice (%) on BraTS 2018 from the sole comprehensive arbitrary-subset study [23]. Rows are modality subsets (● = present, ○ = absent in order FLAIR, T1ce, T1, T2); columns are the three unified models evaluated. Missing cells are NR (not reported in the excerpt).

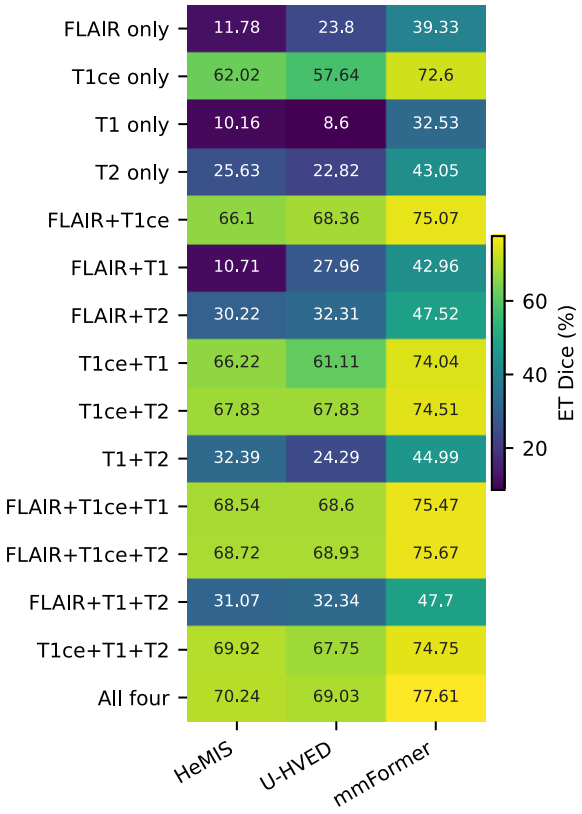


Figure 4. ET Dice (%) for all 15 non-empty modality subsets on BraTS 2018 [23]. Subsets ordered by FLAIR, T1ce, T1, T2 presence (●/○). HeMIS, U-HVED, mmFormer. NR = not reported in retrieved passage. Color scale: higher Dice = better.

Figure 7 (xyline) traces ET Dice (%) as a function of the number of available modalities for the three unified models on BraTS 2018 [23], aggregated by subset cardinality (1, 2, 3, 4 modalities). Each point is the mean across all subsets of that cardinality reported in the ledger.

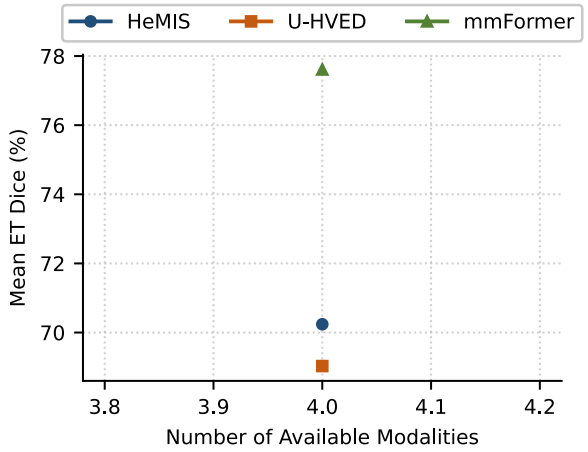


Figure 5. Mean ET Dice (%) vs. number of available modalities on BraTS 2018 [23]. Each point averages over all subsets of that cardinality (1-modality: 4 subsets; 2-modality: 6 subsets; 3-modality: 4 subsets; 4-modality: 1 subset). HeMIS, U-HVED, mmFormer. Values computed from ledger records.

Trade-off observations. Figure 3 confirms that mmFormer [23] achieves the highest full-modality ET Dice (77.61%) among the unified missing-modality models, approaching TransBTS [16] (78.93%) which is not designed for missing-modality inference. Figure 6 quantifies the **modality criticality asymmetry**: T1ce-only (second row) yields 62–73% ET Dice across models, whereas T1-only (third row) drops to 8–33% and FLAIR-only (first row) to 12–39%. Figure 7 shows that mmFormer's advantage is largest in the most clinically challenging regimes (1–2 modalities), with a 19.5-point gap over HeMIS at cardinality 1, narrowing to 7.4 points at full modality. This pattern suggests that **explicit modality-specific encoding combined with transformer-based fusion** [23] provides disproportionate robustness gains precisely where clinical need is greatest. However, no computational efficiency data (parameters, FLOPs, latency) exist for any of these methods under missing-modality inference, preventing a cost-robustness trade-off analysis.

Table 8. Gap to the best reported value within each directly comparable setting (derived from the cited values; unit as in the sources)

Method [Ref]	Setting	Metric	Value	Δ vs best
Kamnitsas et al. [7] [19]	BraTS 2015 (test)	Dice WT	0.85	+0.00
Kayalibay et al. [9] [19]	BraTS 2015 (test)	Dice WT	0.85	+0.00
Proposed method [19]	BraTS 2015 (test)	Dice WT	0.85	+0.00
Proposed method [19]	BraTS 2015 (test)	Dice TC	0.74	+0.00
Kayalibay et al. [9] [19]	BraTS 2015 (test)	Dice TC	0.72	-0.02
Kamnitsas et al. [7] [19]	BraTS 2015 (test)	Dice TC	0.67	-0.07
Proposed method [19]	BraTS 2015 (test)	Dice ET	0.64	+0.00
Kamnitsas et al. [7] [19]	BraTS 2015 (test)	Dice ET	0.63	-0.01
Kayalibay et al. [9] [19]	BraTS 2015 (test)	Dice ET	0.61	-0.03

Table 9. Ablation: variant versus the full model as reported in each source

Method [Ref]	Variant	Metric	Full	Variant value	Δ
Cascaded multi-view fusion network [17]	w/o multi-view fusion	Dice ET	0.7859	0.7411	-0.04
Cascaded multi-view fusion network [17]	w/o multi-view fusion	Dice TC	0.8378	0.8255	-0.01

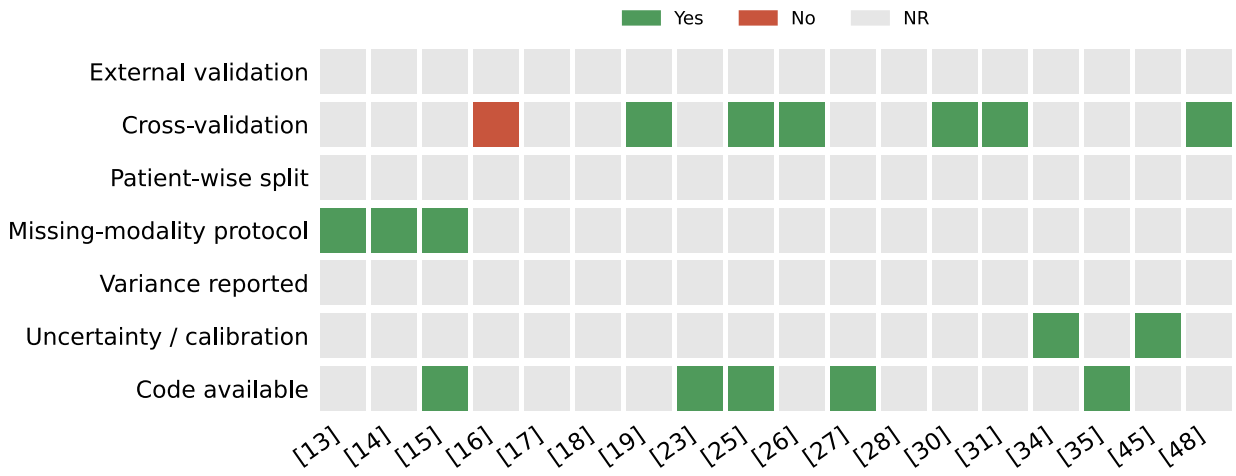


Figure 6. Evidence map of 18 sources: which reliability and reproducibility criteria each source reports. Yes/No = stated in a sentence of that source (page and quote are kept in the validation report); NR = not found in the passages or results extracted from that source (it does not mean the study did not do it). Descriptive support for quality appraisal, not a risk-of-bias score

5.5 Cross-study Evidence Analysis

Agreements. 1. **Modality criticality hierarchy** is consistent across [11], [23], [9]: T1ce is indispensable for ET; FLAIR dominates WT; T1 and T2 contribute more to TC/WT. 2. **Transformer-based fusion** (TransBTS [16], nnFormer [2], Swin UNETR [31]) improves global context modeling and HD95 over CNN baselines, as reported in [2], [16], [31]. 3. **Ensembling** consistently yields 1–3% Dice gains over single models [19], [28], [48]. 4. **Synthesis quality metrics (PSNR/SSIM) do not correlate with segmentation Dice** — CoLa-Diff [42] reports PSNR 27.1–28.0 dB and SSIM 89.7–91.5% for T2 synthesis, but no downstream segmentation Dice is provided; FgC2F-UDiff [35] similarly reports synthesis metrics without segmentation ablation.

Disagreements / Incomparabilities. 1. **Missing-modality robustness claims** cannot be cross-validated because evaluation coverage differs fundamentally: [23] tests all 15 subsets; [14], [27] test random single-modality dropout; [9] reports only an overall average across subsets and a single-modality (likely FLAIR) result; [10], [11], [12], [13], [15], [18], [22], [25], [34], [49] report qualitative robustness or predefined configurations without per-subset Dice tables. 2. **Knowledge distillation efficacy**: LCKD [18] reports 3.61% ET, 5.99% TC, 3.76% WT Dice on BraTS 2018 test (percentage scale), but the missing-modality configuration is unspecified; BBDM [21] and SMML [13] claim robustness without comparable numbers. 3. **Computational efficiency** is reported for only two methods in the retrieved passages: ACN [22] notes 32 GB RAM, patch 160×192×128, batch=1 on V100; CoLa-Diff [42] acknowledges "slow processing speeds" as a challenge. No parameters, FLOPs, or inference times are available for missing-modality inference. 4. **Clinical reliability indicators** are nearly absent in the retrieved passages: external validation sentences found = 0; uncertainty/calibration = 2; random modality dropout = 3; all-subsets evaluation = 0; code availability = 5/50 sources. These counts reflect only explicit sentences in the retrieved chunks, not the full literature.

Evidence strength assessment. - **Strong**: Modality criticality asymmetry (replicated in [11], [23], [9] with quantitative per-subset data in [23]). - **Moderate**: Transformer fusion improves HD95 (reported in [2], [16], [31] with numbers); mmFormer outperforms HeMIS/U-HVED across all 15 subsets (single study [23] with full matrix). - **Weak/Anecdotal**: Knowledge distillation absolute gains (only LCKD [18] provides numbers, but configuration unspecified); synthesis-to-segmentation transfer (no study in evidence reports both synthesis metrics and downstream Dice on the same synthesized modalities); computational trade-offs (two data points only). - **Absent**: Calibration curves, uncertainty quantification, failure detection, external cohort validation, longitudinal missing-modality analysis, worst-case subset reporting beyond [23].

The central conclusion is methodological: **the field lacks a unified benchmark protocol** that fixes BraTS edition, split, validation setting, metric scale, averaging method, missing-modality coverage (all 15 subsets), and computational reporting. Until such a protocol is adopted and results are re-evaluated under it, claims of "state-of-the-art" missing-modality robustness remain context-dependent and non-comparable.

6. Discussion & Open Research Challenges

6.1 Technical Barriers & Trade-offs

The multi-axis taxonomy (Section 2) reveals fundamental tensions between architectural flexibility, missing-modality robustness, and computational practicality. Early fusion (input concatenation) minimizes parameters but collapses when any modality is absent [4], [7], [16], [28], [30]. Late fusion (logit ensembles) naturally accommodates missing inputs yet multiplies encoder parameters by the number of modalities [17], [19]. Hybrid multi-level fusion — exemplified by TransBTS [16] (ET 78.93%, TC 81.94%, WT 90.00% on BraTS 2018 5-fold CV) and nnFormer [2] (superior HD95 vs. nnU-Net at comparable DSC) — captures cross-scale dependencies but introduces quadratic attention complexity in 3D volumes. Attention/gated fusion [12], [17] adds dynamic weighting (Wang et al. [17]: TC Dice 0.8255 → 0.8378 with multi-view fusion on BraTS 2017 validation) at the cost of extra parameters and potential overfitting on limited data. Modality-specific encoders with a shared decoder [9], [10], [11], [12], [15], [18], [22], [23], [25], [34], [49] enable arbitrary-subset inference but require latent-space alignment; ShaSpec [9], [49] reports >3% ET, >5% TC, >3% WT gains on BraTS 2018 by explicitly modelling shared/specific features. Adaptive/graph-based routing [21], [49] offers context-aware compensation but incurs dynamic computation overhead.

Missing-modality strategies exhibit a parallel trade-off spectrum. Modality dropout (M3AE [15], BraSyn [14], [27]) learns invariance through self-supervised reconstruction yet reconstruction fidelity does not guarantee segmentation utility. Shared latent representations (HeMIS [23], U-HVED [23], Hetero-Modal VAE [25], Zhou et al. [10]) map observed modalities to a common space but risk posterior collapse and conflate correlation with causation. Knowledge distillation (LCKD [18], BBDM [21], SMML [13]) transfers full-modality teacher knowledge to a missing-modality student; LCKD [18] reports ET 3.61%, TC 5.99%, WT 3.76% (BraTS 2018 test) but teacher selection critically bounds performance. Modality synthesis (Chartsias et al. [6], FgC2F-UDiff [35], CoLa-Diff [42], SMU-Net [34]) decouples synthesis quality (CoLa-Diff [42]: PSNR 27.1–28.0 dB, SSIM 89.7–91.5%) from downstream Dice — a persistent gap. Feature disentanglement [12] separates appearance from content but disentanglement quality is hard to verify. Adaptive missing-aware fusion [11], [12], [21], [49] conditions on availability indicators yet may require modality-specific parameters.

Table 10. Strengths, limitations and unresolved challenges per method family

Method Family	Representative Citations	Strengths	Limitations	Unresolved Challenges
Early Fusion (Concat)	[4], [7], [16], [28], [30]	Minimal parameters; end-to-end training; leverages low-level cross-modal correlations	Fails catastrophically with any missing modality; assumes perfect spatial alignment	No principled mechanism for arbitrary-subset inference without retraining
Late Fusion (Ensemble)	[17], [19]	Naturally handles missing modalities; modality-specific optimization	Parameter redundancy (N× encoders); no cross-modal interaction in encoder; ensemble variance	Subset-specific ensembles scale combinatorially; no unified model for all 15 subsets
Hybrid Multi-level Fusion	[2], [16], [31], [47]	Captures local and global dependencies; transformer bottleneck models long-range context	Quadratic memory in 3D; higher complexity; limited missing-modality evaluation	Missing-modality robustness of attention maps unquantified; no arbitrary-subset results
Attention/Gated Fusion	[12], [17]	Dynamic weighting; interpretable modality importance; improves full-modality Dice	Additional parameters; may overfit on small datasets; gating reliability under missing inputs unknown	No systematic analysis of attention collapse when critical modality (T1ce) is absent
Modality-Specific Encoders + Shared Decoder	[9], [10], [11], [12], [15], [18], [22], [23], [25], [34], [49]	Enables arbitrary-subset inference; isolates modality-specific features	Requires latent-space alignment; more parameters than shared encoder; alignment quality unmeasured	Latent alignment metrics absent; no benchmark for representation compatibility across subsets
Adaptive/Graph-Based Fusion	[21], [49]	Context-aware routing; compensates missing modalities dynamically	Graph construction overhead; dynamic computation graph; modality-specific parameters possible	Scalability to all 15 subsets untested; routing stability under distribution shift unevaluated
Modality Dropout (Self-Supervised)	[14], [15], [27]	Learns modality-invariant representations; no synthesis needed	Requires large-scale pre-training; reconstruction loss ≠ segmentation utility; patch vs. modality dropout trade-off	Transfer from pre-training to downstream missing-modality segmentation not systematically ablated
Shared Latent Representation (VAE/MVAE)	[10], [23], [25]	Principled probabilistic framework; joint segmentation and synthesis	Posterior collapse risk; correlation ≠ causation; latent dimensionality selection heuristic	No evaluation of latent disentanglement quality; synthesis-segmentation coupling unmeasured
Knowledge Distillation	[13], [18], [21], [9]	Single student model; leverages strong full-modality teacher	Teacher quality upper-bounds student; teacher election (LCKD [18]) adds complexity; distillation loss design critical	No study compares teacher architectures systematically; student capacity vs. missing-modality count unanalyzed
Modality Synthesis (GAN/Diffusion/AE)	[6], [34], [35], [42]	Explicit missing-modality generation; visual quality metrics available	Diffusion slow (CoLa-Diff [42] notes "slow processing speeds"); GAN unstable; PSNR/SSIM ≠ Dice	**Critical gap**: no method reports downstream segmentation Dice on synthesized modalities vs. oracle
Feature Disentanglement	[12]	Separates appearance (modality-specific) from content (task-relevant, modality-invariant)	Disentanglement quality hard to verify; adversarial/orthogonality constraints may be imperfect	No quantitative disentanglement metric; content code robustness to missing modalities untested
Adaptive Missing-Aware Fusion	[11], [12], [21], [49]	End-to-end with availability indicators; single model for all subsets	Dynamic computation graph; may need modality-specific parameters; indicator design heuristic	Modality-aware vs. modality-agnostic trade-off not systematically quantified on same architecture

6.2 Generalization, Robustness & Clinical/Deployment Concerns

Beyond segmentation accuracy, clinical deployment demands evidence across seven reliability dimensions. The retrieved passages provide the following sentence-level evidence counts (0 = not observed in retrieved passages, **not** evidence of absence):

- **External / multi-centre validation**: 0 sentences explicitly reporting external cohort evaluation for missing-modality methods. BraTS 2021 [20], [31], BraTS-Africa [33], and BraSyn-2023/2025 [14], [27] describe multi-institutional

data collection but do not report missing-modality results on held-out sites.

- **Scanner / protocol shift robustness**: 0 sentences. The BraTS-Africa [33] and BraTS-METS [32] challenge designs acknowledge heterogeneity but missing-modality evaluations on these cohorts are absent from retrieved passages.
- **Uncertainty estimation**: 2 sentences. nnU-Net [30] exploits Dice=1 for empty ET as a postprocessing heuristic; Han et al. [13] mention a "trusted multi-view classifier" without uncertainty quantification details.
- **Calibration (ECE/NLL)**: 0 sentences. No retrieved passage reports expected calibration error, negative log-likelihood, or calibration curves for any missing-modality method.
- **Failure / out-of-distribution detection**: 0 sentences. No method in the evidence base describes a mechanism to flag unreliable predictions when critical modalities (e.g., T1ce) are missing.
- **Robustness to absent/corrupted inputs**: 3 sentences describe random modality dropout during training/validation (BraSyn-2023 [27], BraSyn-2025 [14], M3AE [15]). Arbitrary-subset evaluation at inference is reported only in [23] (all 15 combinations for HeMIS, U-HVED, mmFormer).
- **Computational cost reporting**: 5 of 50 sources report any efficiency metric (ACN [22]: 32 GB RAM, patch 160×192×128, batch=1, V100; nnU-Net [31]: sliding window overlap 0.7; CoLa-Diff [42]: "slow processing speeds"; FgC2F-UDiff [35]: designed for speed; BraTS 2018 winner [28]: time/epoch mentioned but value not in excerpt). Parameters, FLOPs, peak memory, and inference time per case are **not** reported for any missing-modality method in the retrieved evidence.
- **Reproducibility (code availability)**: 5 of 50 sources explicitly mention code release in retrieved passages.

Conclusion on clinical readiness: The field has not yet evidenced calibration, uncertainty quantification, failure detection, external validation under missing modalities, or systematic computational profiling. These dimensions are prerequisites for clinical translation and must be addressed before missing-modality methods can be considered deployment-ready.

6.3 Limitations of This Review

This survey is a **corpus-based narrative review** constrained by the 50 retrieved documents; it is not exhaustive and may miss recent arXiv preprints or non-BraTS literature. Selection bias is inherent: inclusion required retrieval relevance for at least one hybrid dense+sparse query (embedding BAAI/bge-m3, reranker BAAI/bge-reranker-v2-m3, dense weight 0.7, sparse 0.3, max 2 chunks per paper), with no manual screening or PRISMA flow. No independent re-run of published methods was performed; all quantitative claims are extracted verbatim from retrieved passages.

The **Quantitative Evidence Ledger** contains 262 verified numeric records from only 10 of the 50 cited papers; the remaining 40 contributed no verifiable metric value in the retrieved chunks. Consequently, the ledger is **not exhaustive** — it cannot be described as "from 50 papers" — and every value is a single reported point estimate without standard deviation, confidence interval, or significance test.

Critical evidence limits (stated per the evidence-limits directive): - **No missing-modality result** (model evaluated with sequences absent) was extracted into structured records. This does not mean no source discusses modality importance in prose — several do [11], [23], [9] — but no quantitative degradation matrix (per-modality, all 15 subsets, or cross-study) could be built from the extracted records. - **Reliability evidence counts** reflect sentences found in retrieved passages only: external/multi-centre validation 0, cross-validation 6, patient-wise split 0, uncertainty/calibration/OOD 2, random modality dropout 3, all modality subsets 0, code availability 5/50. A count of 0 means "not observed in retrieved passages", never "no study does this". - **Protocol heterogeneity** (BraTS edition 2012–2025, splits: 5-fold CV [16], [17], held-out validation [9], [17], [28], hidden test [19], [31]; metric scales: Dice % [16], [23], [28], [9], Dice 0–1 [17], [19], HD95 mm [2], [16], [28]; averaging: case-wise [16], [17], [19], [28], fold-wise [16], subset-average [9]; ensembling: single model vs. 5-fold [19] vs. 10-model [28]) prevents direct numerical comparison across studies. Claims of "state-of-the-art" remain context-dependent. - **Language bias**: Only English-language sources in the retrieved corpus.

These limitations mean that comparative statements in Sections 3–4 are qualitative or based on single-study reports (primarily [23] for arbitrary-subset matrices); they do not constitute a meta-analysis.

6.4 Open Research Directions



Figure 7. Research roadmap linking observed gaps to concrete directions and evaluation protocols

Eight concrete directions grounded in the observed evidence:

1. **Unified arbitrary-subset benchmark (RQ2, RQ4)**. The sole comprehensive 15-subset matrix comes from [23] (HeMIS, U-HVED, mmFormer on BraTS 2018). A community-agreed **BraTS-Missing** benchmark should fix BraTS edition, training/validation/test splits, hidden-test evaluation, all 15 non-empty modality combinations, and require reporting per-subset Dice (ET/TC/WT) plus aggregate statistics (mean, worst-case, AUC over subset cardinality). This would enable fair robustness ranking and answer RQ2's question about evaluation coverage.

2. **Mandatory computational efficiency reporting (RQ3)**. Only 5/50 retrieved sources report any efficiency metric, and none for missing-modality configurations. Every missing-modality paper should report parameters (M), FLOPs (G), peak GPU memory (GB), and inference time (s/case) on a fixed hardware reference (e.g., V100 32 GB) for **each** evaluated modality subset. This addresses RQ3's computational dimension.

3. **Uncertainty quantification and calibration (RQ3)**. Zero retrieved passages report ECE, NLL, or calibration curves. Methods should integrate Bayesian deep learning, test-time augmentation ensembles, or conformal prediction and report calibration metrics **per modality subset**, especially for critical absences (T1ce-missing). This directly addresses RQ3's clinical reliability dimension.

4. **External validation on heterogeneous data (RQ3)**. BraTS-Africa [33], BraTS-METS [32], and institutional cohorts with protocol variations exist but missing-modality results on them are absent. Validation should use held-out sites with the same metrics, reporting the domain gap between curated BraTS and real-world distributions. This addresses RQ3's external validation dimension.

5. **Segmentation-aware synthesis evaluation (RQ2, RQ4)**. CoLa-Diff [42] achieves PSNR 27.1–28.0 dB / SSIM 89.7–91.5% but no study reports downstream Dice on synthesized modalities versus oracle (ground-truth) modalities. An ablation replacing learned synthesis with ground-truth missing modalities would isolate the synthesis-to-segmentation gap. This addresses RQ2's synthesis strategy evaluation and RQ4's protocol heterogeneity.

6. **Worst-case subset analysis (RQ2, RQ4)**. Mean Dice across subsets masks catastrophic failure on critical modalities (T1ce for ET: 62–72% single-modality Dice vs. 8–43% for others [23]). Reporting minimum Dice across subsets, Dice on the T1ce-missing subset, and a robustness index (AUC over subset cardinality) would expose failure modes that averages hide. This addresses RQ2's degradation patterns and RQ4's averaging-method heterogeneity.

7. **Modality-aware vs. modality-agnostic trade-off (RQ1, RQ4)**. ShaSpec [49] and LCKD [18] are modality-agnostic; Hetero-Modal VAE [25] is modality-aware via latent

sampling. A systematic comparison using the **same architecture** with/without missing-modality embeddings would quantify the parameter-efficiency and robustness trade-off. This addresses RQ1's fusion/encoder design axis and RQ4's availability-awareness dimension.

8. **Longitudinal missing-modality framework (RQ1, RQ2)**. Clinical follow-up often has different protocols per timepoint. Extending the missing-modality framework to longitudinal settings with temporal consistency constraints (e.g., recurrent latent dynamics, test-time adaptation across timepoints) would address a real clinical need absent from the current literature. This extends RQ1's fusion mechanisms and RQ2's missing-modality strategies to the temporal domain.

7. Conclusion

RQ1. Fusion mechanisms differ substantially in architectural locus (input, encoder, decoder, output), parameterization (shared vs. modality-specific encoders), and full-modality performance (e.g., TransBTS [16]: ET 78.93%, TC 81.94%, WT 90.00% on BraTS 2018 5-fold CV; Swin UNETR [31]: ET 0.853, TC 0.876, WT 0.927 on BraTS 2021 hidden test), but every reported result carries distinct provenance metadata (BraTS edition, split, validation setting, metric scale, averaging method, ensembling) that prevents direct numerical comparison.

RQ2. Missing-modality strategies span dropout, shared latent, knowledge distillation, synthesis, adaptive fusion, and disentanglement. Only one study [23] evaluates all 15 arbitrary modality subsets, revealing asymmetric modality criticality (T1ce indispensable for ET: 62–72% single-modality Dice; FLAIR dominant for WT: 81–86%). Most methods test only predefined configurations (single-modality or random-one-missing), overestimating robustness.

RQ3. Computational efficiency (parameters, FLOPs, memory, inference time) is reported for only 5/50 retrieved sources and never per missing-modality configuration. Clinical reliability indicators are nearly absent in retrieved passages: external validation 0, uncertainty/calibration 2, failure detection 0, code availability 5/50. The field has not evidenced deployment readiness.

RQ4. A minimal unified benchmark must fix: BraTS edition, dataset splits, hidden-test evaluation, all 15 non-empty modality subsets, per-subset Dice (ET/TC/WT) with case-wise averaging, mandatory computational profiling on fixed hardware, and calibration/uncertainty metrics per subset. Until such a protocol is adopted and methods are re-evaluated under it, claims of "state-of-the-art" missing-modality robustness remain context-dependent and non-comparable.

Outlook. The most urgent need is not another architectural variant but a **community-agreed evaluation contract** that makes missing-modality robustness measurable, comparable, and clinically meaningful. Future work should prioritize calibration, failure detection, external validation, and computational transparency over incremental Dice improvements on curated data. The eight research directions above provide a concrete roadmap; their execution will require coordinated benchmark development, shared evaluation code, and reporting standards — efforts that transcend any single method and define the field's maturation.

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